

Seedling Production

The production of healthy and uniform transplants is the first step towards having a successful crop so it must be carried out with great care. To achieve this, make sure that the containers (trays and pots) are filled evenly, verify the quality of the soil and provide careful and regular maintenance.

Each week, the propagation calendar is consulted to see which crops and cultivars need to be started.

In the nursery charts, you will find all the details related to transplant production for each crop - the number of days in plug cells, the number of seeds per cell, the type of potting mix, the optimal germination temperature, the type of container and the number of days before potting up.

Choice of Potting Mix

To produce vigorous seedlings for a diversified farm, you need at least 2 types of potting mixes. One is for starting most seeds, and the other is for potting up.

- Low fertility potting mix does not contain compost, has a low salinity and has a fine texture. These characteristics favor seed germination.
- High fertility potting mix contains around 15% compost and is geared towards plant growth.

We use two types of commercial potting mixes. Make sure that all the ingredients in the substrate you plan to use are permitted by your organic certifier (some wetting agents are not approved).

Although it is possible to buy the ingredients for the potting mix separately and make your own recipe, it is much easier to buy the mixes commercially. Buying commercial potting mix ensures consistent quality and requires less time. It is also possible to add mycorrhizae to the potting mix (or buy an already inoculated one) as a strategy to inoculate your fields.

Choice and Maintenance of Containers

The standard size for nursery trays is 10 in (25 cm) by 20 in (50 cm) and they should be perforated under each plug cell to ensure proper drainage.

These black plastic trays are available in several models and qualities. For the sake of standardization and efficiency, we choose to limit the types of containers we use:

- 128 cell trays
- 72 cell trays
- 4 in (10 cm) pots
- 6 in (15 cm) pots
- Open trays, 2.5 in (5 cm) high (to transport pots)
- Open trays 1.25 in (3.2 cm) high (microgreens)

After transplanting, the trays and pots are shaken out and the remaining potting mix is left in the field. This potting soil is not reused, as it has been leached of its nutrients. The empty containers are brought back to the nursery tables. They are rinsed and left on the tables for a few days to solarize. The clean containers are then stacked for storage. This is much simpler and less labor intensive than disinfecting them individually with sanitizers.

Do not leave empty containers in the sun for too long as this accelerates their deterioration. Properly maintained, trays and pots can be used for about 3-4 years.

Filling the Trays

First, add water to the soil. This is done using large plastic tubs on wheels. Hydrating the peat moss in the potting mix will increase its porosity. For an 85 liter bag, add about 8 liters of water. Mix the water in with a small shovel until the moisture is evenly distributed. To check if the potting soil contains the right amount of water, take a handful and squeeze it in your fist. You should see some water between your fingers, but you should not be able to get it to drip. For greater efficiency, always mix 2 bags at a time.

- To fill the trays, place 4 of them in front of you and pour about 4 liters of potting mix over each of them. Spread the potting soil evenly over the 4 trays, without pressing down, breaking up clumps if necessary.

- One by one, lift the trays 6 inches and let them fall flat on the workbench. Remove the excess with your hands, without applying pressure, and make sure that all the cells, especially those on the edges and in the corners, are evenly filled.
- Stack the trays 4 by 4 and press the top of the stack so that a small depression is created in the center of each plug cell where the seed will be placed. The top tray will be done last.
- The trays are now ready to be seeded.

Seeding

A fundamental element of efficient transplant production is identifying the right seeding method according to the type of seed. Most round seeds, as well as pelleted seeds, such as Salanova lettuce seeds, can be seeded with a vacuum seeder. However, non-pelleted lettuce seeds are tiny and should be sown by hand. For other types of seeds that have an irregular shape, hand seeding is unavoidable. There are small hand-held seeders that can make the task easier.

Vacuum Seeder

For all seeds that are compatible (see nursery chart), we use a vacuum seeder. This speeds up seeding in trays a great deal, allowing us to seed up to 120 trays per hour with great precision. There are plates for the various types of seeds and trays, we use those that are compatible with 128 cell and 72 cell trays.

Place the trays on the work bench in front of you in stacks of 4 or 5. The seeder is on your left. Clean the seeder plate with a brush to remove dust; this will increase the suction power and allow for better seed adhesion.

- Pour seeds onto the plate. Use enough to seed a few trays, but not too much as this will slow down the process. Plug in the vacuum, turn it on and open the suction valve.

- To distribute the seeds on the plate, move the seeder from front to back and from left to right, while shaking it slightly on each side, so that the seeds are always in motion.
- When all the holes in the plate contain seeds, remove the excess with your fingers so each hole contains only one. Make sure that all excess seeds end up in the gutter.
- Run your hand quickly over the seed tray to remove excess soil.
- Place the seeder over a tray (seeds facing toward the tray) and close the suction valve. Gently tap on the seeder to make sure all the seeds have fallen out. Do not open the suction valve again until you have removed the seeder from the tray, otherwise soil will be sucked in and will stick to the plate.
- Brush the plate clean and repeat.

Note that it is possible to build your own vacuum seeder. However it is not recommended, as it is a precision tool.

The holes in the plates are drilled on order, depending on the model and brand of your trays. To make the most of your vacuum seeder, limit the number of models and brands of trays you use.

Finalizing the Seeding

Place the trays in stacks of 4 or 5 as you finish seeding them, crisscrossing them at a 45-degree angle to prevent the seeds from sticking to the tray above. Once the seeding is done, take the trays to the benches and cover the seeds with a small amount of potting mix to keep them moist.

Trays should be organized on benches by crop, cultivar and transplanting date to facilitate their transport to the field or hardening off area.

Water all the trays at the same time at the end of a seeding session.

Depending on the optimal germination temperature and the value of the seeds, it may be worthwhile to use plastic domes over the trays or to place them on heating mats until seeds germinate.

Watering

Watering the trays is done in two steps. First, to wet the surface of the soil, then to allow water to flow throughout the entire cell.

The total amount of water needed depends on many factors, including temperature, relative humidity and sunlight. Dry, sunny weather means more frequent watering, while wet, cloudy weather means watering is drastically reduced, and sometimes even unnecessary.

In our context, in the spring, irrigation is carried out between 10 am and 2 pm. During the summer months, it is carried out between 9 am and 4 pm.

Quick Tips for Nursery Irrigation:

- Inspect the surface and underside of the tray to see if it is dry.
- Lift the tray to assess its weight. The drier the tray, the lighter it is.
- When possible, remove a plug from the tray to see how much moisture it contains. It is sometimes necessary to add more water than you think. That said, be careful not to overwater, as this leaches nutrients from the soil, impacts seedling growth and can lead to fungal problems such as damping off.
- In general, the older the crop, the more water it will need. However, the frequency will be reduced.
- In general, the younger the crop, the more often it is irrigated. However, less water is needed.
- Irrigation is an art that takes some time and a lot of observation to master; only one person should be in charge in order to become familiar with this crucial step in the production of beautiful transplants.

During Growth

Irrigation of the containers should be uniform at all stages of growth.

Adjust the pressure of the watering hose while pointing it next to the containers (on the ground). The pressure should be intermediate: enough to reach the furthest trays, while not splashing the soil. Then, water the trays with a sweeping motion, taking care to go past each side and overlapping each pass so as to give the same amount of water to each cell.

Irrigation Regime in Nurseries According to the Stage of the Crop:

Stage 1: Germination – spray nozzle 1,000 micro holes

- Frequent watering, especially in sunny conditions.
- Avoid humidity variations.
- Keep the surface of the soil moist.
- Do not allow to dry.

Stage 2: Young seedling - spray nozzle 400 micro holes

- Water as needed.
- Letting the surface of the cells dry is beneficial to the young seedlings. It encourages the rootlet to grow and fully colonize the soil.

Stage 3: Mature transplant - spray nozzle 400 micro holes

- Less frequent but more abundant watering.
- Let the soil dry before watering again.
- Prepare transplants for outdoor conditions.
- Avoid the proliferation of algae on the surface of the soil.

Stage 4: Mature transplant in pots - remove the spray nozzle

- Less frequent but more abundant watering.
- Water by putting the water directly into the pot, avoiding the foliage.

- Fill the pots with water, then let the soil absorb it. Repeat as needed.

Allow the potting mix to dry before watering again to prepare the plants for outdoor conditions and to avoid algae growth on the soil surface.

Fertilization

The choice of potting mix is very important to ensure that the plants have everything they need during their stay in the nursery. For particularly demanding crops or those with a history of deficiencies, it is possible to systematize the application of fish hydrolysate before the appearance of symptoms. The use of fish hydrolysate or other soluble fertilizers can also be done after the appearance of deficiency symptoms. When using fertilizer, it is very important to read the label carefully and to respect the recommended dosage.

For plants in 6 inch (15 cm) and 4 inch (10 cm) pots that have a major deficiency, compost, pelleted chicken manure or other solid fertilizer can be added directly to the pots, being very careful to avoid direct contact with the roots.

For open trays or cell trays, you can prevent deficiencies by placing them over compost bins while they await transplantation; the roots will draw the nutrients they need from the compost.

Potting Up

Some crops that spend more time in the nursery need to be potted up (refer to the nursery chart). Potting up refers to transplanting a plant into a larger container and often into more fertile soil. Potting up gives the young plants more space and nutrients, thus promoting their development. Producing more developed and robust transplants also promotes their rapid recovery after being transplanted and shortens the time between transplanting and the first harvest.

We typically only choose to pot up greenhouse crops that have a high commercial value and/or a long life cycle, such as tomatoes, eggplants, peppers and ground cherries. We also pot up crops that have a low or uneven germination rate, such as celery.

Here are the steps to follow for effective potting up:

- Put on some good music.
- Properly irrigate the trays to be potted up.
- Moisten the new soil.
- Fill the pots to the brim with potting mix and then tamp it down with another pot. The level of potting mix, once packed, should be 2.5 cm (1 in) below the level of the pot.
- Place the newly filled pots in open transport trays.
- Prepare a few trays before potting up.
- As with transplanting, pull the plug from its initial cell and place it in its new pot. You can do several plugs at a time.
- Place the plants in the new potting mix, in the center of the pot. Note that for grafted crops, you must be very careful not to bury the graft line. For other crops, you can generally bury the plug up to its first true leaves (not cucurbits), which encourages the development of the root system.
- Pot up one cultivar at a time and ensure that they are properly labeled on the new trays.
- The trays containing the new pots are placed on the nursery benches and watered generously. The 400 micro hole spray nozzle can be used for the first few days, then it is better to water directly into the pots (see irrigation instructions stage 4).
- When the leaves of the plants start to touch (1-2 weeks later), plant density should be lowered. Remove every second pot and place them in new trays.

Plant Protection

Firstly, it is important to understand that vigorous and healthy plants are more resistant to insect and disease pressure. Every effort should be made to produce such plants.

According to the principles of integrated pest management, all problems are then treated with the same method. The steps, in order, are the following:

- Avoid introducing pests into the system by making sure to use healthy inputs (seeds, purchased plants, potting mix, fertilizers and water).
- Prevent organisms already present on the farm from entering and growing in the nursery.
 - Consider the use of insect netting
 - Seal holes
 - Avoid unnecessary travel between the greenhouses and the nursery
 - Wash hands before handling plants
 - Disinfect tools
 - Solarize or disinfect trays before using again
 - Let the soil dry between watering
 - Develop and maintain biodiversity on the farm
 - Use ultrasonic rodent repellents
 - Disinfect the nursery once a year
- Regular scouting and identification.
 - Examine plants, trays (for plant damage and presence of pests), at least once a week.
 - Use yellow sticky traps to detect the presence of flying insects
 - Identify the organisms that are observed
- Implementation of mechanical control methods.
 - Installation of sticky traps
 - Installation of mouse traps (or putting sticky traps around the legs of tables to prevent them from climbing)
 - More aggressive irrigation can dislodge aphids from plants and slow their reproduction.

- Scrape and remove algae to slow it down
- Preventive integration of beneficial organisms.
 - Beneficial insects
 - Inoculation of mycorrhizae
 - Inoculation of beneficial fungi (RootShield™)
 - Cats can be useful to control rodent populations
- Curative integration of beneficial organisms.
- Use of biopesticides once the threshold for intervention is reached.

The main challenges of plant protection in nurseries are the following:

- Aphids attack all the crops, especially greens and solanaceous plants.
- Thrips are found in cucurbits and onions.
- Whiteflies love tomato and eggplant plants.
- Rodents love newly-sprouted spinach and bell pepper seeds, but are also a hazard to many other crops.
- Algae grow on soil that does not dry frequently enough.
- Sciarids develop mainly in the moist potting soil of trays that remain in the nursery for a long time (crops that are potted up, celery, celeriac, parsley, flowers and cucumbers).
- Damping off (Pythium) is a fungal disease that can be very damaging. Depending on the history and pressure, beneficial fungus inoculants (Rootshield) can be used as a preventive or curative measure.

Pruning/Pinching/Thinning

Some crops need to be pruned in the nursery. Here is a summary:

- Onions, green onions and shallots are trimmed back about 1 inch (2.5 cm) a week before transplanting to encourage robust growth.

- The head of greenhouse tomato seedlings is pinched so that they can develop two heads.
- It may be useful to take advantage of the watering and scouting rounds in order to thin out plug cells where too many seeds have germinated.

Climate Management

Technically, each crop thrives in slightly different climatic conditions. In a diversified farming context, we aim for the average conditions to be between 64°F (18°C) and 70°F (21°C) which all crops can adapt to. Keep the temperature the same during the day and at night.

Insufficient light in relation to the ambient temperature will lead to plants becoming etiolated.

Relative humidity should be 65-80% for transplant production. Since the number of plants is relatively small in the nursery, it can be difficult to keep the humidity level this high. You can increase the humidity level in the nursery artificially with the help of a misting system. Some growers water the floor on hot, sunny days for the same reason.

Hardening Off

With the exception of plants that will be transplanted in a heated greenhouse, hardening off seedlings is done one week before the transplant date.

Hardening off consists of getting the plants used to outdoor conditions (by exposing them to direct light, wind, temperature variations). Ideally, the plants are hardened off in an intermediate shelter, to reduce the shock. This step is important to encourage a healthy recovery in the field.

Our hardening off greenhouse is adjacent to the nursery in order to limit transport time when moving trays each week.

Transplantation

Before transplanting, the trays or pots should be watered thoroughly. Watering in two waves ensures that the water completely penetrates and moistens all of the substrate.

It may be beneficial to delay transplanting for a few days if the seedlings' roots have not colonized the entire cell by the scheduled transplantation date, and if it is difficult to remove the plugs from their containers without having them crumble.

Similarly, some trays can be planted earlier if growing conditions have accelerated maturity. The quality of a transplant degrades rapidly due to the limited volume of substrate. The roots of a plug that is too old curl up in the bottom of the cell and it becomes difficult to take it out. The shock from transplanting will also be more significant for these seedlings.

In any case, it is important to keep a record of the ideal number of days the plants should stay in their cells. These can indeed vary according to the season and climate management. Reread these notes at the end of the season and adjust the transplant production chart according to your situation.

Once the transplanting is done, the beds should be watered deeply once more to reduce the shock and ensure the seedlings recovery quickly.

Task Management in the Nursery

In order to ensure the smooth operation of the nursery, only one person should be in charge. The responsibilities of this person should include:

- Watering the trays
- Climate management
- Pest scouting and management
- Ordering equipment and inputs
- Follow-up of weekly tasks and management of priorities: seeding, potting up, moving containers to hardening off area, cleaning up

- Keeping the workplace organized

Every other Monday morning , the person in charge of the nursery should:

- Extract important information from the production schedule:
 - Which seedlings are scheduled for the current week, including type of potting mix, type of container, number of seeds per cell
 - Scheduled transplants
 - Scheduled potting up: new container, new potting mix
 - Scheduled hardening off
- Set aside seeds to be used.
- Move the trays that will be transplanted the following week to the hardening off area.
- Other maintenance or supply tasks.

These tasks, and all the information needed to perform them, should be readily available and accessible to the people who will be doing them. The necessary equipment, inputs and tools should be available in the right quantities and be ready to use to facilitate work. If these tasks are delegated, care must be taken to ensure that they are carried out correctly.